

EXACT MONITOR v2.0 - FIRMWARE TIMING

Created by Jordan Korenzvit, April 4, 2022



DOCUMENT STATUS **DRAFT**
DOCUMENT VERSION 0.1
DOCUMENT OWNER Jordan Korenzvit

This document describes when and how loggers and relays log data, transmit data and communicate with one another and the cloud.

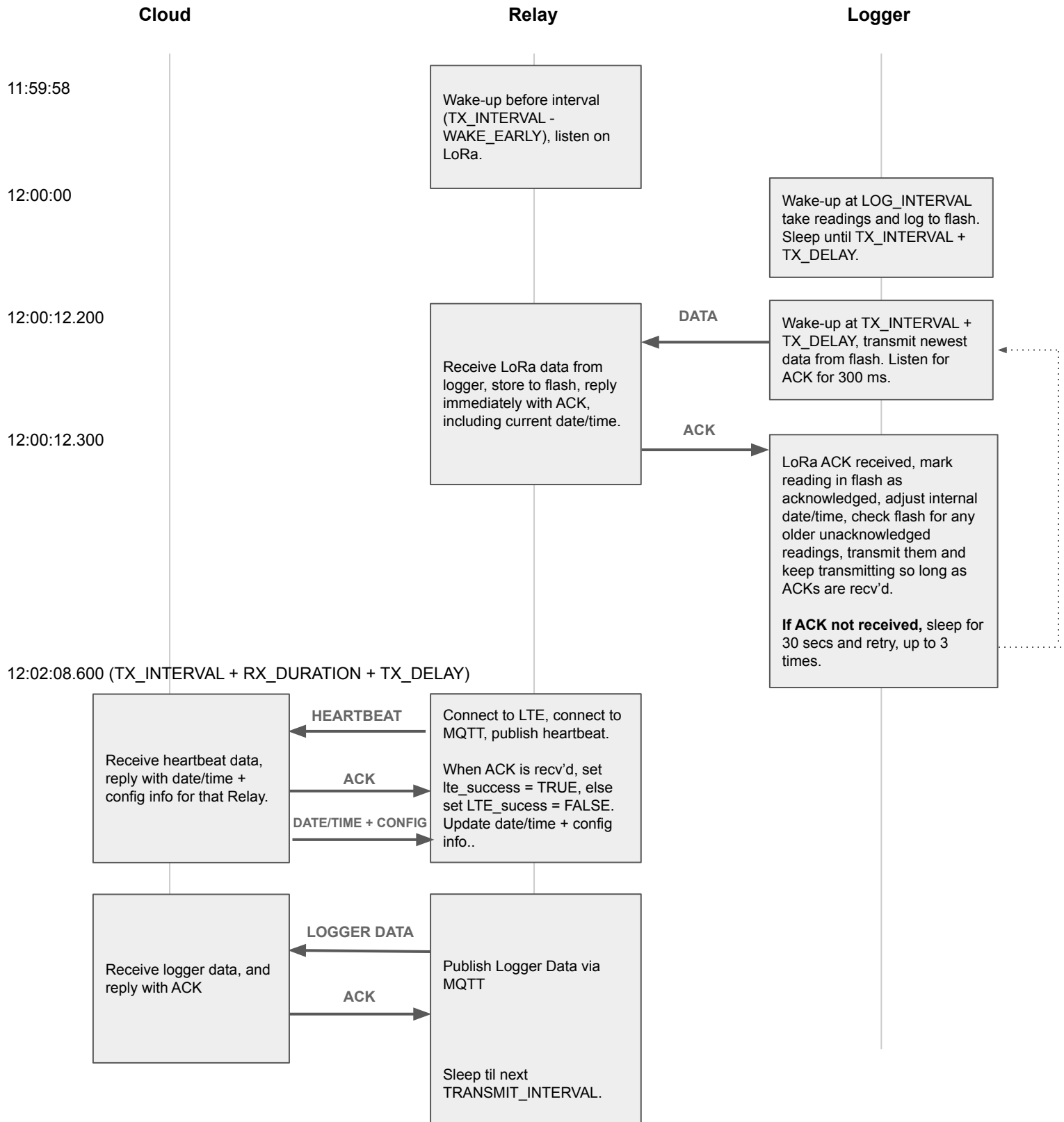
VARIABLES

DEVICE_MODE (0 or 1)	0 = Relay. Listen for LoRa packets and forward to cloud. Also logs and transmit any locally connected probes via Heartbeat. 1 = Logger. Logs and transmits local probes.
RADIO_MODE (0 or 1)	Only applicable when DEVICE_MODE = 0 (Logger). 0 = LTE 1 = EXACT LoRa 2 = BLE only <i>*Review advertising interval, etc.</i> 3 = LoRaWAN <i>*future</i> <i>** future: perhaps each mode can be on/off, and prioritized.</i>
POWER_MODE (0 or 1)	0 = Power Saver. Sleep between intervals. 1 = Always On. RELAY: Always listening to LoRa, connected to LTE and subscribed to MQTT to receive commands. LOGGER: Always connected to LTE and subscribed to MQTT to receive commands. Default is 0. When connected to power, default is 1.
LOG_INTERVAL (secs)	Frequency logger takes readings, and stores to flash. Interval is always synched to the clock. E.g. interval of 900 will log readings at 12:00:00, 12:15:00, 12:30:00, etc. Interval of 60 at 12:00:00, 12:01:00, 12:02:00, etc. Min = 60 / Max = 900.
LOG_INTERVAL_ALARM (secs)	Frequency logger logs readings, when an ALARM_THRESHOLD is crossed. E.g. any temperature > 60°C. Min = 60 / Max = 900.
TX_INTERVAL (secs)	Frequency logger or relay transmits data to via LoRa or LTE. Min = 60. Max = 86,400 (24 hours)
TX_INTERVAL_ALARM (secs)	Frequency logger or relay transmits, under alarm condition. Min = 60. Max = 86,400 (24 hours)
WAKE_EARLY (secs)	Default = 2 seconds. How much earlier than transmit_interval a Relay wakes up to listen for loggers. Helps catch logger w/clocks that are slightly fast. Min = 0. Max = 840. Only valid for DEVICE_MODE = 0.
TX_DELAY (ms)	# of ms after TX_INTERVAL, for transmitting data via LoRa or LTE. Minimizes LoRa collisions and MQTT server load. For a logger, random value between 0 and 29500, rounded to closest 200 ms. Random value determined at first transmit, then saved if transit was successful, resets again if ACK not immediately received.
RX_DURATION (secs)	How long Relay stays in RX mode. Default 120 secs. Only valid for DEVICE_MODE = 0.
ALARM_THRESHOLD (units vary)	For a temperature logger, "60" means "60°C" If any reading on a logger is > or < (depending on ALARM_DIR) than ALARM_THRESHOLD, logger will remain in alarm mode until the threshold is crossed again. Units vary as this can also be used for relative humidity %, PSI, PSF, etc. depending on type of logger.
ALARM_DIR (">" or "<")	E.g. ">" when any logged value is greater than ALARM_THRESHOLD. "<" when any logged value is less than ALARM_VALUE.

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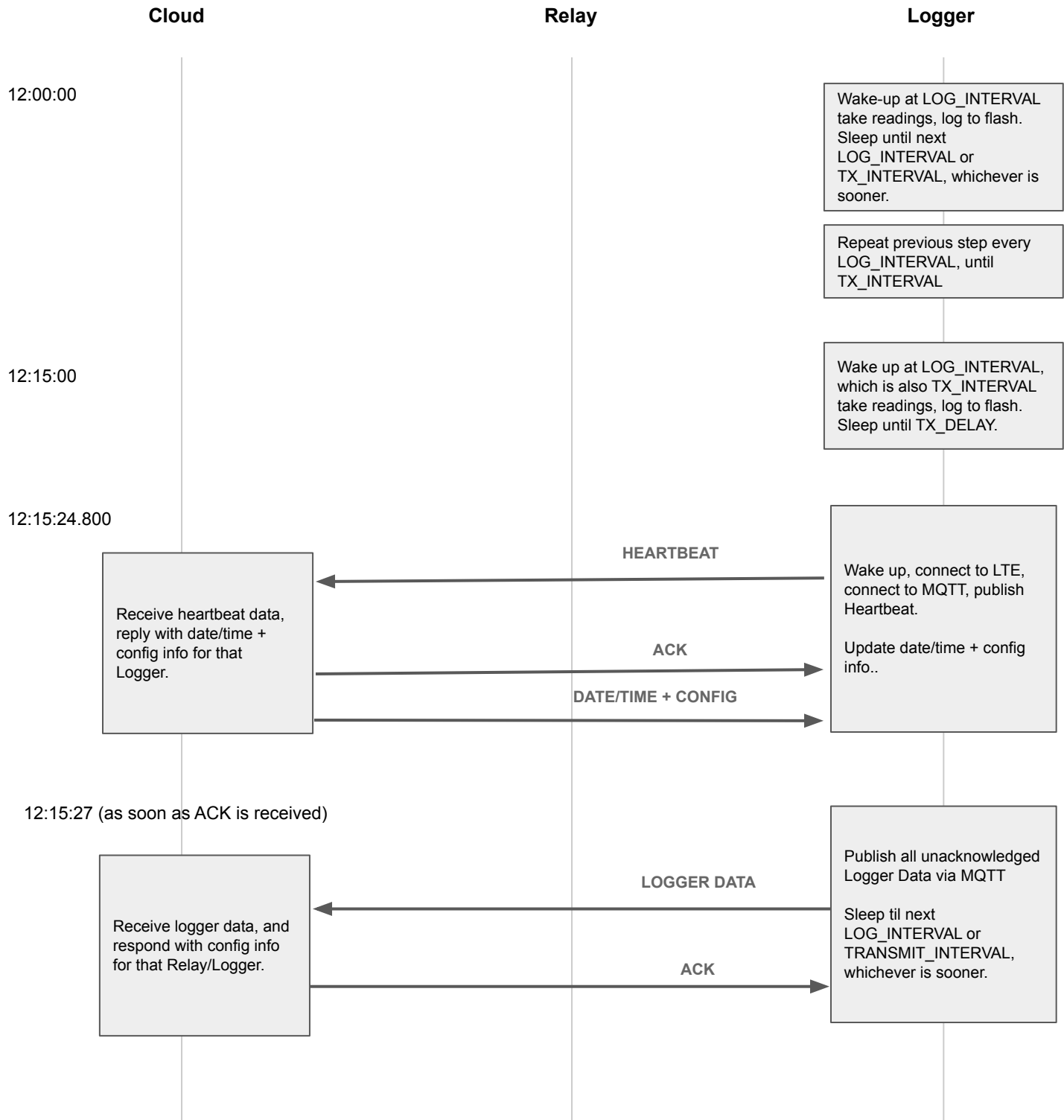
SCENARIO: RELAY + LORA LOGGER



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SCENARIO: LTE LOGGER, LOG_INTERVAL = 60, TX_INTERVAL = 900, TX_DELAY= 24800



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OTHER CONSIDERATIONS

PARENT FUNCTION	LoRa Loggers currently have a function so only the “parent” Relay sends an ACK back to the Logger. This prevents multiple Relays from replying with ACKs simultaneously causing LoRa collisions. When a logger first transmits, it flags its parent with an “O” (for orphan). Any Relay that receives data from an orphan logger, will send back an ACK with its 4-digit Relay-ID (previously last 4-characters of ParticleID). The logger saves the Relay-ID to flash, and uses this next time it transmits data. If the Parent Relay doesn't ACK the logger after 3 TX retries, the logger turns into an orphan, and creates a new random TX_DELAY.
PACKET #	We used this on loggers to track how many connection/tx attempts it needed before data hit a relay and got through to cloud. From 0 through 99. At 99 it reset to 0 again. It would change to -1 if the device restarted for some reason, e.g. watchdog.
BACK-OFF TIMING	If device can't connect to LTE, how many retries and how often? Retries should get spaced further and further apart so that the device doesn't kill its battery when no connection is available. LoRa could back off too.
LTE FALLBACK	If RADIO_MODE = 1 (LoRa), should device fallback to LTE after certain period of time? 24 hours? 7 days? 1-month? Programmable? Could be a variable . E.g. # of hours, or 0 for never.
HEARTBEAT DATA	Cellular signal, cellular carrier, battery level, date/time. Should it include: temperature data from onboard inputs, or should this go to diff MQTT channel?
LOGGER CONFIG OVER LORA	We currently send some logger config data back with LoRa ACK from Relay. E.g. TX_INTERVAL, a command to reclaim saved data. Should we continue with this? BLE could replace this, except it wouldn't work unattended.
SHIPPING MODE	Sleep a bunch more when we ship it. Plugging in a probe could activate regular mode. Might not be necessary with BACK-OFF TIMING.